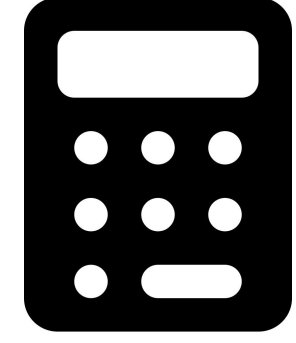


Motivation

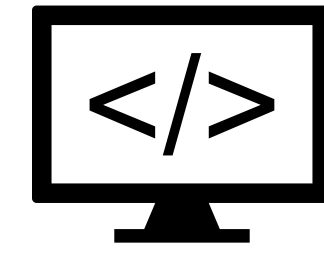
Tool use is widely considered an important characteristic of both natural and artificial intelligence. Cognitive scientists study both tool use and tool discovery; the ability to use a tool is often distinct from the ability to invent new tools. While LLM tool use is now ubiquitous, little is known about LLM tool *discovery*.



Web search



Computation & Data analysis



Development & Coding assistants

To study discovery, we need a novel setting that allows us to observe progress towards each aspect of tool creation. Almost all modern benchmarks mandate tool creation and suffer from training data contamination. We develop a small parameterized family of environments that allows us to decompose tool creation into its constituent parts.

Lomekwi

Task: Open N locked **doors**, each of which requires a distinct **key**, within a certain **budget** of turns.

Actions: **Investigate** a door (option to pick up a random key and/or a random shard); **Use** an item on a door; **Combine** two items.

Mechanics: N doors/keys and T shards. Exactly two of the shards combine into the **machine**, and no other items combine. The machine is reusable and creates the corresponding key when used on a door. *The agent does not know the machine exists, and all names are obfuscated.*

Choice: Either hunt for keys one at a time (slow “coupon collector”), or spend time trying to combine shards into a tool (the machine).

Analysis: Brute-force coupon collector costs $\mathbb{E}[\text{grind}] = NH_N + N$ turns, where H_N denotes the N th harmonic number. Discovering, building, and using the machine costs $\mathbb{E}[\text{build}] = \mathbb{E}[\text{collect}] + \mathbb{E}[\text{combine}] + \mathbb{E}[\text{open}]$, where

$$\mathbb{E}[\text{collect}] = TH_T, \quad \mathbb{E}[\text{combine}] = \frac{\binom{T}{2} + 1}{2}, \quad \mathbb{E}[\text{open}] \approx N \left(1 + \left(1 - \frac{1}{N} \right) \mathbb{E}[\text{collect}] \right).$$

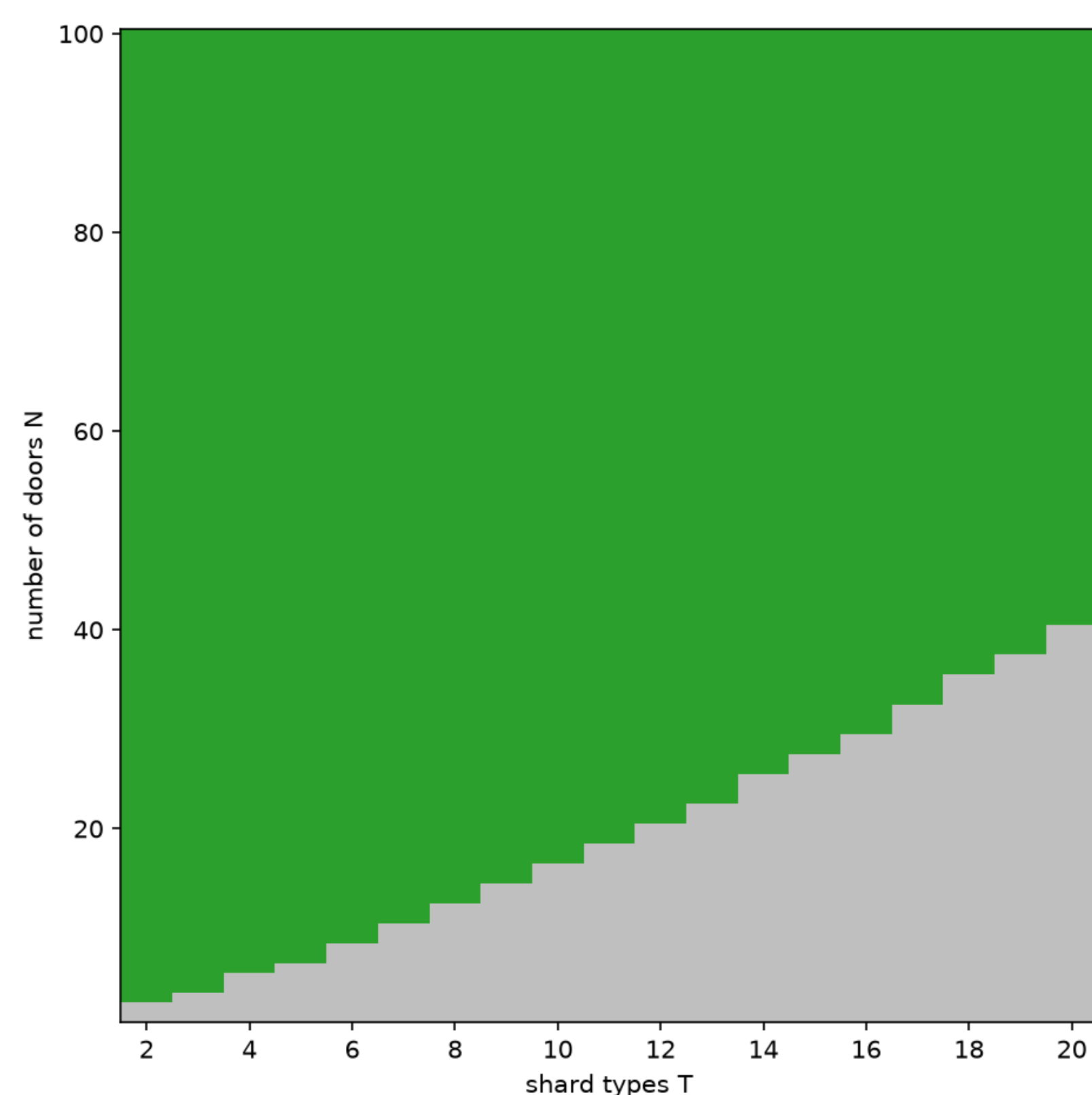


Figure 1. Expected actions to solve, building vs. grinding over a task region (T shard types $\times$ N doors). Green indicates that $\mathbb{E}[\text{build}] < \mathbb{E}[\text{grind}]$. Building is cheaper than grinding almost everywhere.

Three Steps of Tool Discovery

Tool-use benchmarks report only the **success rate**. But discovering a reusable tool and solving a task with it requires three steps in order:

1. *gathering* the parts the tool needs,
2. *building* the tool, and
3. *using* the tool to finish the task.

Since each step is conditional on the previous, the total probability of success $S = \Pr(\text{solve})$ splits exactly into these steps:

$$S = \underbrace{C}_{\text{curiosity}} \cdot \underbrace{R}_{\text{recognition}} \cdot \underbrace{E}_{\text{efficiency}} + \underbrace{G}_{\text{no tool built}}$$

- **Curiosity** $C = \Pr(\text{hold})$ — does the model collect the parts?
- **Recognition** $R = \Pr(\text{build} \mid \text{hold})$ — *given that it holds the parts*, does it build?
- **Efficiency** $E = \Pr(\text{win} \mid \text{build})$ — *once built*, does it complete the task within budget?
- **Grind** G — solved via brute force without building the tool.

Results

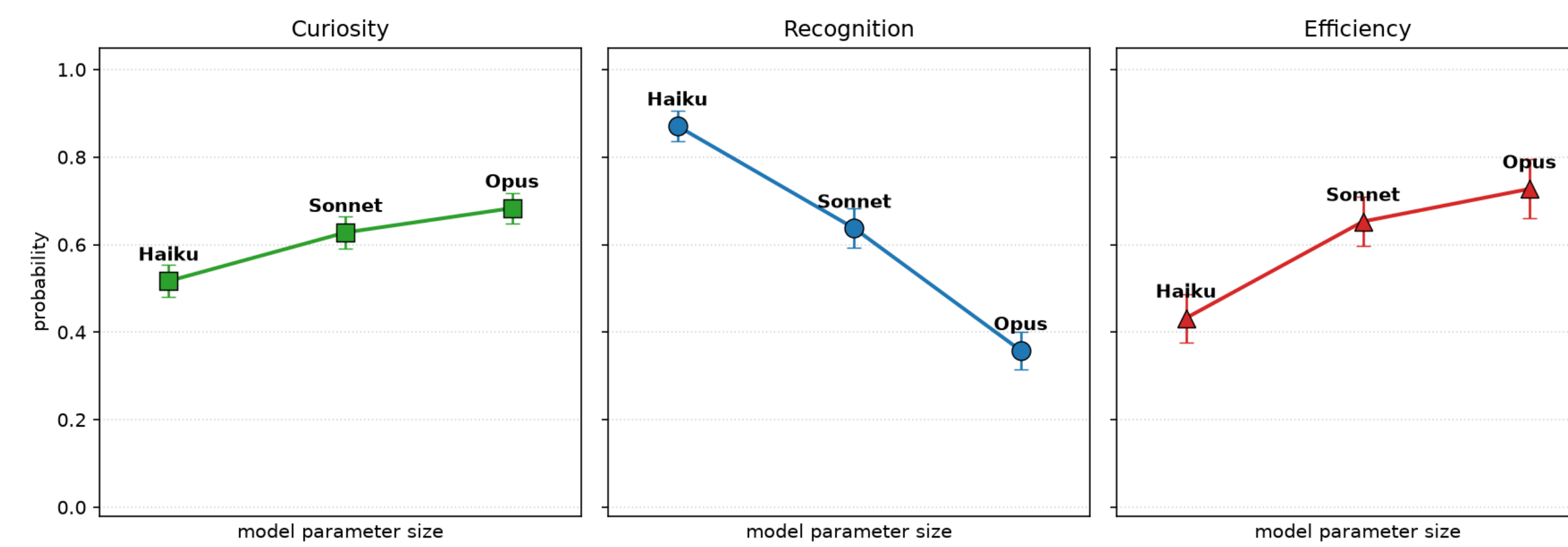


Figure 2. Tool discovery metrics for Anthropic models on Lomekwi, 180 runs per.

Figure 2 displays the results of a parameter sweep over $2 \leq T \leq 10$ and $N \leq 20$ (total 180 runs per model). **Curiosity** (left) is relatively high and uniform across models. **Efficiency** (right) improves with size. But **recognition** (middle) *drops* at the frontier: Haiku 0.87 $\rightarrow$ Sonnet 0.64 $\rightarrow$ Opus 0.36. **Bigger models hold the correct parts and still grind instead of building.**

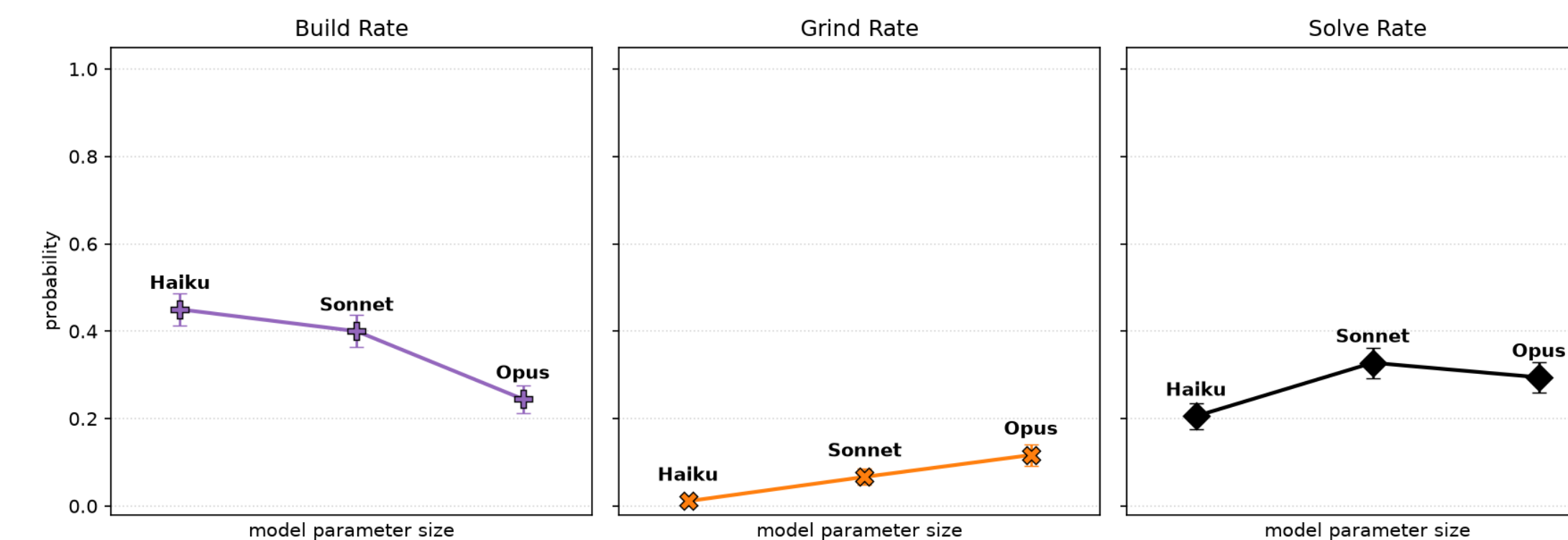


Figure 3. Overall metrics for Anthropic models on Lomekwi, 180 runs per.

As shown in Figure 3, overall metrics hide the details in model behavior. Better grinding ability with capability masks the falling build rate.

Sweep across difficulties

Figure 4 displays the complete data from the Lomekwi parameter sweep.

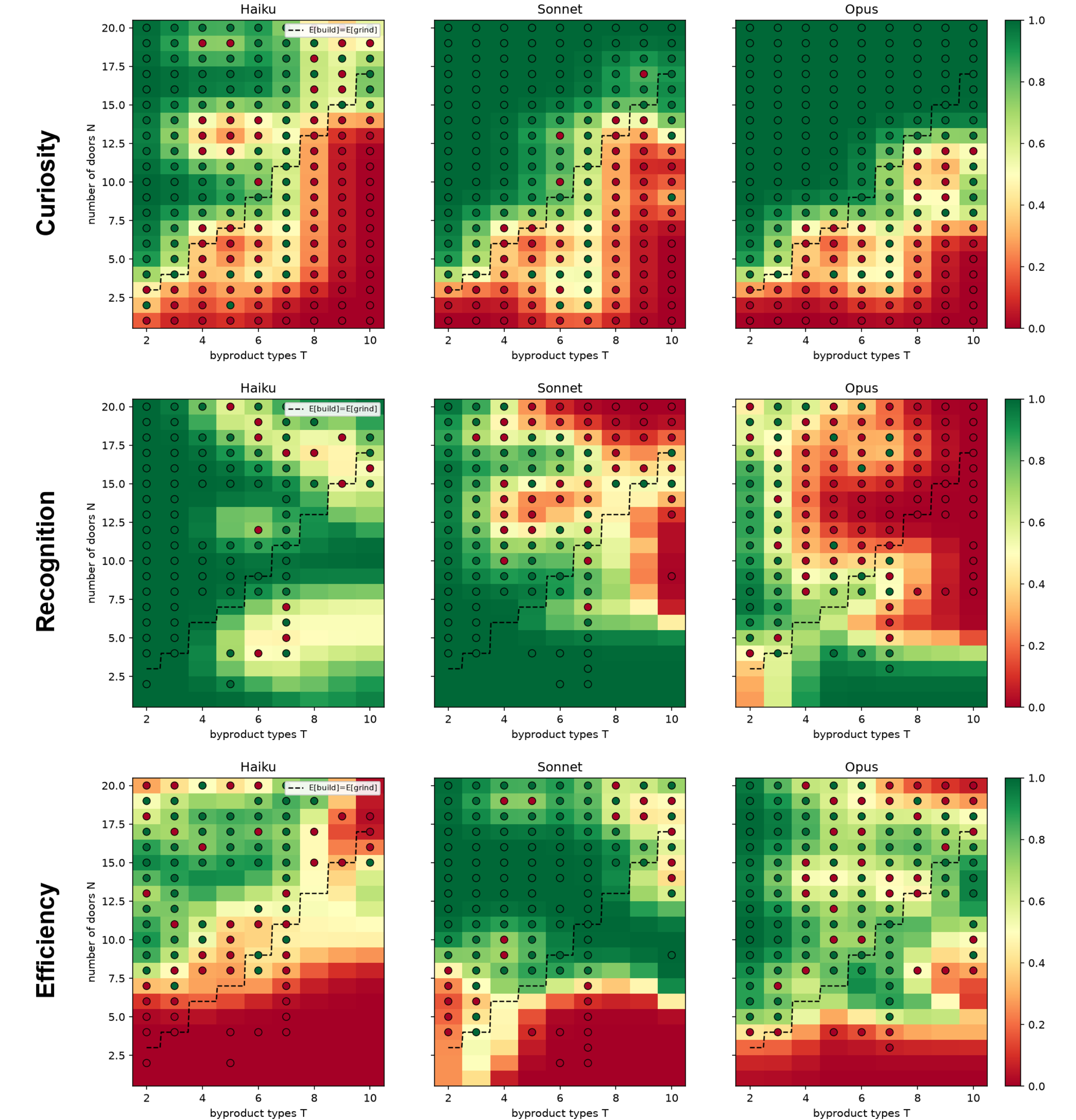


Figure 4. The complete data from the Lomekwi sweep. Each cell is one setting (N doors $\times$ T part-types). Color represents the value of the metric (green high, red low), with Gaussian pooling. Columns: Haiku, Sonnet, Opus. Optimal build-grind boundary shown as a dotted line.

Contributions

Our experiments suggest that decomposing tool creation exposes significant differences within model families. One might expect stronger models to be both better tool users and tool *inventors*. However, our results show that larger models start exploring tool opportunities less frequently. Our primary contributions:

1. A **decomposition of tool discovery** into individually measurable parts (recognition, curiosity, efficiency, and grind),
2. A **controlled, prior-suppressed environment** (Lomekwi) in which LLM behavior can be compared against an optimal policy, and
3. An **inverse scaling result** showing that recognition decreases with model size in this environment.

In addition, we note the following two aspects of Lomekwi that are crucial to observing tool discovery.

- **Obfuscated** object names, so a model can’t rely on pretrained priors.
- **Budgeted** actions so grinding is possible but risky.